

HSIU-CHI LU

Ph.D. Student, Network Science Institute, Northeastern University, Boston, MA, USA

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Research Interests

Computational Social Science, Network Science, Complex Systems, Relational Sociology, Polarization, Information Diffusion, Digital Media

Education

Northeastern University, Network Science Institute, Boston, USA Sep. 2025 – Present

Doctor of Philosophy in **Network Science**

Advisor: David Lazer

National Chengchi University, Taipei, Taiwan Aug. 2022 – Aug. 2024

Master of Science in **Sociology**

Advisor: Hsuan-Wei Lee, Jen-Hao Chen

Thesis: *Agents of Discord: Modeling the Impact of Social Bots on Opinion Polarization in Complex Networks*

National Chengchi University, Taipei, Taiwan Aug. 2018 – Jul. 2022

Bachelor of Science in **Sociology**

Minor in **Applied Mathematics**

Publications

Peer-Reviewed Journal Articles

- Jen-Hao Chen* & **Hsiu-Chi Lu*** (2026). “Evolving Landscapes, Shifting Narratives: Understanding Taiwanese News Media Portrayals of LGBTQ+ Populations from 2010 to 2021.” *Journal of Homosexuality*. [Link](#).

*Equal authorship.

- Hsiu-Chi Lu** & Hsuan-Wei Lee (2026). “Modeling Competing Narratives in Adaptive Networks: How Social Pressure and Network Dynamics Drive Tipping and Persistence.” *Humanities and Social Sciences Communications*. [Link](#).
- Hsiu-Chi Lu** & Hsuan-Wei Lee (2024). “Agents of Discord: Modeling the Impact of Political Bots on Opinion Polarization in Social Networks.” *Social Science Computer Review*. [Link](#).

Conference Presentations

- “Culture War Without Partisan Consolidation: Public Lifestyle Signaling in America.” (2026). Oral presentation at the **Politics and Computational Social Science Conference (PaCSS 2026)**, APSA Political Communication Preconference, Boston, MA, Sep. 2026.
- “Culture War Without Partisan Consolidation: Public Lifestyle Signaling in America.” (2026). Oral presentation in the special session “**Structure and Meaning: Linking Social Networks and Cultural Life**” at the **2026 Annual Meeting of the American Sociological Association (ASA)**, New York, NY, Aug. 2026.
- “Culture War Without Partisan Consolidation: Public Lifestyle Signaling in America.” (2026). Oral presentation at the **12th International Conference on Computational Social Science (IC2S2 2026)**, Burlington, VT, Jul. 2026.
- “The Stage and the Closet: Stigma and the Distortion of Digital Traces of Pornography.” (2026). Poster presented at the **International School and Conference on Network Science (NetSci 2026)**, Boston, MA, Jun. 2026.
- “Agents of Discord: Modeling the Impact of Political Bots on Opinion Polarization in Social Networks.” (2024). **Sunbelt 2024**, International Network for Social Network Analysis (INSNA), Edinburgh, Scotland, Jun. 2024.
- “Modeling Disinformation Resilience: How Social Pressure and Network Dynamics Drive and Sustain the Spread of False Narratives.” (2024). Poster presented at the **10th International Conference on Computational Social Science (IC2S2 2024)**, Philadelphia, PA, Jul. 2024. [Link](#).
- “A Journey of Inclusiveness: Mapping Emotions and Phrases for LGBTQ in Major Taiwanese News Outlets over the Past Decade.” (2024). Poster presented at the **10th International Conference on Computational Social Science (IC2S2 2024)**, Philadelphia, PA, Jul. 2024. [Link](#).
- “Analysis and Reflection on the Relationship and Actions Between AI and Humans in the Information Age: A Case Study of GPT-3 and its Creations.” (2021). **2021 Annual Meeting of the Taiwanese Sociological Association**, Taipei, Taiwan, Nov. 2021. (Presented in Mandarin). [Link](#).

Selected Undergraduate Research Project funded by the National Science and Technology Council (NSTC).

Academic Experience

Research Assistant

Network Science Institute, Northeastern University

Aug. 2025 – Present

Boston, MA, U.S.A.

- Conducted independent computational social science research using large-scale digital trace and panel data to investigate cultural landscapes, lifestyle polarization, and digital data biases.
- Processed, analyzed, and visualized complex datasets from the National Internet Observatory (NIO), synthesizing empirical findings into comprehensive technical reports.

Research Assistant

College of Health, Lehigh University

Aug. 2024 – Aug. 2025
Bethlehem, PA, U.S.A.

- Developed a model to investigate disinformation spread in online social networks, focusing on mechanisms that lead to widespread dissemination; resulting in a peer-reviewed journal article in *Humanities and Social Sciences Communications*.
- Implemented mathematical functions and agent-based simulations to model narrative diffusion, tipping dynamics, and persistence in adaptive social networks.

Research Assistant

Department of Sociology, National Chengchi University

Feb. 2022 – Aug. 2025
Taipei, Taiwan

- Conducted independent and collaborative research on LGBTQ representation in major Taiwanese news outlets during the 2010s, resulting in a conference presentation and a peer-reviewed journal article in *Journal of Homosexuality*.
- Developed a structured database of approximately 20,000 records by utilizing Python-based web crawling to collect over 50,000 news articles from major Taiwanese media outlets. Employed natural language processing, machine learning, and large language models to filter and classify data.
- Applied statistical methods to analyze patterns between sexual orientation and external factors, investigating societal inclusiveness and enhancing clarity through data visualization.
- Analyzed night work patterns and their relationship with marital status among Taiwanese individuals, contributing insights to a work-life balance report; additionally, used Airtest to monitor mobile devices and gather data on college students' mental health.

Research Assistant

Institute of Sociology, Academia Sinica

Nov. 2022 – Jul. 2024
Taipei, Taiwan

- Conducted research on digital society issues such as misinformation, online polarization, political bots, and narrative diffusion, resulting in peer-reviewed publications.
- Designed and visualized agent-based simulations in Python to examine the interaction between political bots, polarization, and the spread of disinformation.
- Analyzed network data to explore the role of network structure in opinion diffusion and disinformation dissemination.

Lab Intern

Institute of Information Science, Academia Sinica

Jul. 2023 – Dec. 2023
Taipei, Taiwan

Lab: **Natural Language Processing and Sentiment Analysis Lab**

- Supported the laboratory team in crafting a questionnaire for experiments on cognition of misinformation and processing demographic data from respondents.
- Contributed to interdisciplinary collaborations by integrating social science expertise into AI research endeavors.

Data Scientist

Big Data Group, Taiwan Institute of Governance and Communication Research

Feb. 2022 – Jun. 2023
Taipei, Taiwan

- Developed Python-based web-crawling scripts to gather social media data, creating three databases from Facebook and Ptt (a Taiwanese Reddit-like forum). The codes are available on [here](#), and the datasets can be requested via this [link](#).
- Structured and analyzed social media data using natural language processing and statistical methods to explore public opinion on the pandemic and the 2022 local elections in Taiwan.

Teaching Experience

Teaching Assistant

National Chengchi University.

Sep. 2022 – Feb. 2023

Course: **Data Analysis in Sociology**

- Prepared and conducted a one-hour discussion class on R and statistics each week.
- Supported over 70 students in group settings with their final papers, assisting them in discussing research questions, solving statistical problems, and providing programming support.

Awards and Grants

- **Master Dissertation Award.** Taiwanese Sociological Association 2024
- **Travel Grants.** 10th International Conference on Computational Social Science 2024
- **Outstanding Student Overseas Conference Grant.** Foundation for the Advancement of Outstanding Scholarship 2024
- **Subsidy for Students Attending International Conference.** Ministry of Science and Technology, Taiwan 2024

- **Undergraduate Research Project Scholarship.** Ministry of Science and Technology, Taiwan 2020
Research Topic: Analysis and Reflection on the Relationship and Actions Between AI and Humans in the Information Age: A Case Study of GPT-3 and its Creations
- **Presidential Award,** National Chengchi University 2019